



Final Examination
Automatic Image Analysis
Summer term 2017

Computer Vision &
Remote Sensing

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Name:

Student ID number:

Auxiliary resources: none

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July 18, 2017

DO NOT OPEN THIS EXAMINATION SHEET UNTIL YOU ARE TOLD TO DO SO!

Write your **name** and **student ID** in the corresponding places at the top of this page **now**.

Books, notes, dictionaries, own empty sheets of paper, and pocket calculators are **not allowed**.
Use only a pen. Everything written with a pencil will not be taken into account.

If you do not understand a question, please **ask**.
It will be to your advantage to read the entire examination before beginning to work.

The exam is a **multiple choice** test.

For each question there is at least **one and at most four** correct answers.

The number of points p for a single correct answer are stated next to the question.

Please note, that there is a **penalty of $-p/2$ points** for a wrong answer.

The minimal number of points for each question is 0 (i.e. no negative points for whole questions).

	Which of the following numbers is even?				2P
	i) 2	ii) 3	iii) 4	iv) 5	
Example 1			X (correct +2P)		Result: 2P
Example 2		X (incorrect: -1P)	X (correct +2P)		Result: 1P
Example 3	X (correct +2P)		X (correct +2P)		Result: 4P
Example 4		X (incorrect: -1P)			Result: 0P

Notation:

Black = Gray level of 0 ; White = Gray level of 255

Image gradients point from dark to bright.

Angles are defined counter-clockwise.

Lots of luck and do your best!

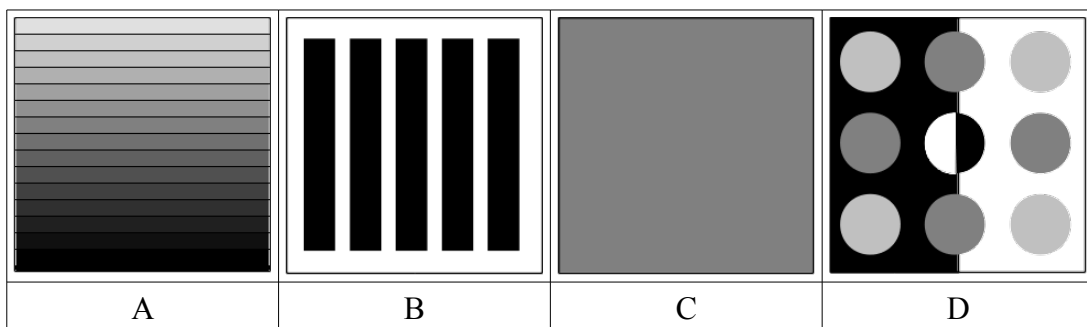
Block I

1. In the context of image analysis, what is usually referred to as “**Gaussian scale space**”? 1P

- i) A set of images, which are - starting from the original image - consecutively **smoothed with a Gaussian kernel** of increasing size.
- ii) An **image pyramid**, in which the scale of each level is sampled from a Gaussian distribution.
- iii) The application of a filter bank consisting of **Gaussian kernels** as well as the **first and second derivative** of a Gaussian function at different scales.
- iv) The assumption, that the scale (i.e. the **size**) of **objects** in images follows a Gaussian distribution.

2. A common measure for the information content of an image patch is the **entropy** of the corresponding grey-scale distribution. The **grey-scale distribution** of an image patch describes the relative occurrence of intensity values within the patch. 2P

If the following image patches are sorted according to **ascending entropy** values of their corresponding grey-scale distribution, which is the right order?



- | | | | |
|------------|-------------|--------------|-------------|
| i) D-C-B-A | ii) A-B-C-D | iii) A-D-B-C | iv) C-B-D-A |
|------------|-------------|--------------|-------------|

Block II

3. The **probability density** function of a one-dimensional Gaussian distribution over a continuous variable x is defined as $p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$. 1P

If $\mu=0, \sigma=1$, then the **probability** $P(x=0)$ is

i) $1/\sqrt{2\pi}$	ii) 0	iii) $\sqrt{2\pi}$	iv) 1
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4. An event A can take two possible values, i.e. a_1 and a_2 . A second event B can take two possible values b_1 and b_2 . Both events are **independent**. It is known that: $P(A=a_1) = 0.2$, $P(A=a_2) = 0.8$, $P(B=b_1) = 0.5$, $P(B=b_2) = 0.5$. 2P

Which of the following is the correct **joint probability** $P(A=a, B=b)$?

$P(A=a, B=b)$	$a=a_1$	$a=a_2$	$P(A=a, B=b)$	$a=a_1$	$a=a_2$
$b=b_1$	0.1	0.4	$b=b_1$	0.25	0.25
$b=b_2$	0.25	0.25	$b=b_2$	0.25	0.25
i)			ii)		
$P(A=a, B=b)$	$a=a_1$	$a=a_2$	$P(A=a, B=b)$	$a=a_1$	$a=a_2$
$b=b_1$	0.1	0.4	$b=b_1$	0.1	0.1
$b=b_2$	0.1	0.4	$b=b_2$	0.4	0.4
iii)			iv)		

5. Let $p(x) = p(x_1, x_2, x_3)$ be the **joint probability density** of three variables x_1, x_2, x_3 . Which of the following operations is an example of **marginalization**? 1P

i) $p(x_2, x_3) = \int p(x) dx_1$	ii) $p(x_1) = \int \int p(x) dx_2 dx_3$
iii) $p(x) = p(x_1 x_2, x_3) p(x_2 x_3) p(x_3)$	iv) $p(x) = p(x_1) p(x_2) p(x_3)$

6. $P(C=c | X=x) = \frac{P(X=x | C=c) P(C=c)}{P(X=x)}$ 1P

The term $P(C=c | X=x)$ in the equation above is the

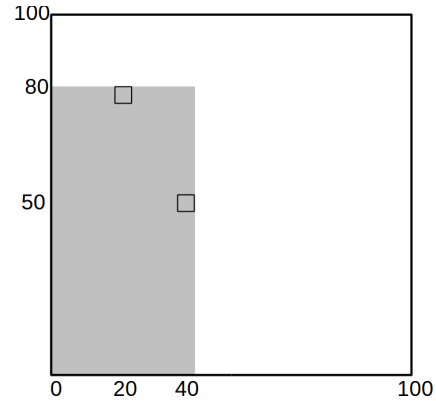
i) likelihood of X .	ii) posterior of C .	iii) likelihood of C .	iv) posterior of X .
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7. A typical image analysis task is to distinguish between the **background and the foreground** object. A certain **probabilistic framework** provides the probability $p(x|C=c)$ of observing certain features x given either foreground $C=1$ or background $C=0$. Within this application scenario, it is known that the foreground covers on average around **10%** of the image area. The final label of a pixel shall be decided by an **MAP estimator**. A possible **decision rule** in this case is: 1P

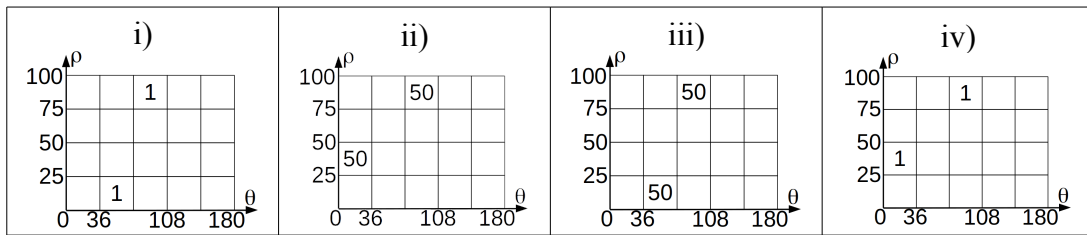
- i) Label x as background if $P(C=0) > P(C=1)$, otherwise as foreground.
- ii) Label x as background if $\log(P(x|C=0)) - \log(P(x|C=1)) > \log(0.1) - \log(0.9)$, otherwise as foreground.
- iii) Label x as background if $9 \cdot P(x|C=0) > P(x|C=1)$, otherwise as foreground.
- iv) Label x as background if $P(x|C=0) > P(x|C=1)$, otherwise as foreground.

Block III

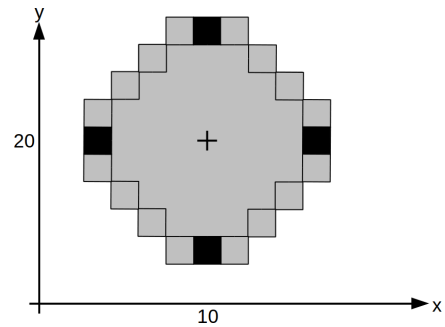
8. The figure on the right shows a 100×100 pixel large image, with a white (grey value 255) and grey (grey value 205) area. Two distinct pixels are marked as black squares. A **Hough line detector** (parametrized as in the lecture) is applied to this image. If only the two marked pixels are taken into account, which of the following is the correct **voting space**?



2P



9. The figure on the right shows a simple shape in grey, where **boundary pixels** are denoted with black-outlined squares. Which of the following **R-tables** should be used to look for an instance of this object that is scaled by $\sqrt{2}$ and counter-clockwise rotated by 45° . **Only the black-filled pixels are taken into account.**



3P

Please note: $\cos(45^\circ) = \sin(45^\circ) = 1/\sqrt{2}$

	i)	ii)	iii)	iv)
[0-45)				
[45-90)	[0, 4]	[-4, -4]	[-4, 4]	[-4, 0]
[90-135)				
[135-180)	[4, 0]	[4, -4]	[4, 4]	[0, 4]
[180-225)				
[225-270)	[0, -4]	[4, 4]	[4, -4]	[4, 0]
[270-315)				
[315-360)	[-4, 0]	[-4, 4]	[-4, -4]	[0, -4]

10. The **ISM** aims at detecting objects in images and is very similar to the Generalized Hough Transform. However, instead of an R-Table, it uses a **codebook**. This codebook contains

1P

- The **gradient orientation** of edges within the object.
- Prototypes** for object parts with **stable appearance** in different images.
- The **similarity of object parts** to each other.
- Offset vectors** that point to a reference point on the object.

Block IV

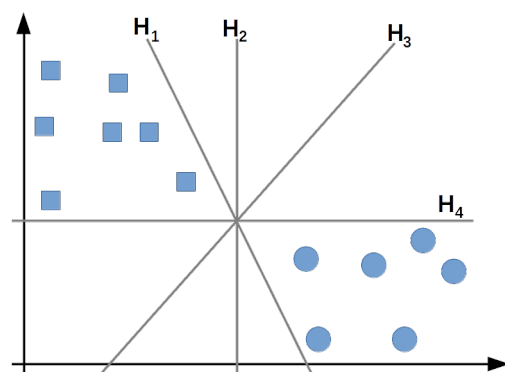
11. **Bag of Visual Words (BoVW)** refers to one of the standard and more traditional approaches of image analysis. It aims to describe the image / object as a whole which then can be classified. 1P

Which of the following algorithms describes the **general procedure** of BoVW in the context of image analysis?

i) 1. Apply a keypoint detector to the image. 2. Cluster the keypoints based on a suitable description. 3. Assign offset vectors to the found clusters which point to a reference point on the object. 4. During prediction each keypoint votes using all offset vectors of the cluster to which it was assigned.	ii) 1. Apply a keypoint detector to the image. 2. Cluster the keypoints based on a suitable description. 3. For each query image, count how many of the detected keypoints have been assigned to each cluster. 4. Compare the calculated histogram to the histograms in the database.
iii) 1. Apply a gradient operator to image and template. 2. Compute the local correlation between both gradient images. 3. Find positions where the local correlation exceeds a threshold. 4. These positions correspond to object positions within the image.	iv) 1. Divide the image into a grid of non-overlapping blocks. 2. For each block compute a local histogram of gradient orientations. 3. Concatenate all histograms to a final descriptor. 4. Compare the descriptor of the query image to the descriptors in the database.

12. The figure on the right shows different **decision boundaries** between data of two classes (denoted by squares and circles). Which of these hyperplanes does most likely belong to a linear SVM?

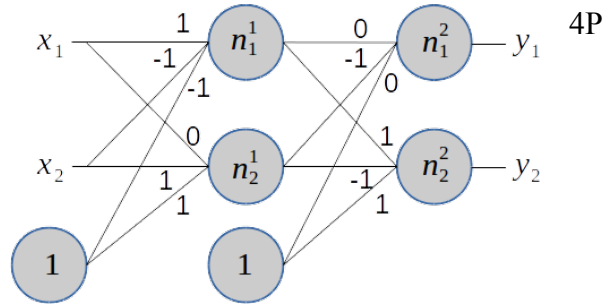
- i) H_1 ii) H_2
 iii) H_3 iv) H_4



1P

Block V

13. The **MLP** shown in the figure on the right consists of three layers, with two neurons n in all layers. The **bias term** is denoted as an additional neuron in each layer with 1 as constant output. Within the figure, the numerical **weight** values are attached to the corresponding connections. A **ReLU** is used as activation function in all layers (besides the input layer): $\phi(z) = \max(0, z)$.



The training of the MLP is based on computing the **gradient** of an error function with respect to the weights. A typical choice is the squared Euclidean norm, i.e.

$e = 0.5 \|y^L(x_\alpha) - y_\alpha\|_2^2$ which compares the desired output y_α with the actual output $y^L(x_\alpha)$ of the network.

Given the current state of the network and an **input** $x_\alpha = (x_1, x_2)_\alpha = (2, 1)$ with $y_\alpha = (y_1, y_2)_\alpha = (0, 1)$, what is the value of the error e ?

i) $1/\sqrt{2}$	ii) $\sqrt{2}$	iii) 1	iv) 1/2
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14. Given N samples, which of the following update rules refers to **offline learning**? 1P

- i) $\Delta w_{ij}^l = -\eta \frac{\partial e}{\partial w_{ij}^l} - \eta \lambda w_{ij}^l$ ii) $\Delta w_{ij}^l = -\eta \frac{\partial e}{\partial w_{ij}^l}$
 iii) $\Delta w_{ij}^l = -\eta \sum_{\alpha=1}^N \frac{\partial e}{\partial w_{ij}^l}$ iv) $\Delta w_{ij}^l(t+1) = -(1-\beta) \eta \delta_j o_i + \beta \Delta w_{ij}^l(t)$

15. In comparison to standard MLPs, **ConvNets** usually 1P

i) use weights shared among units.	ii) have more layers.	iii) use handcrafted features.	iv) have units with receptive fields.
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Block VI

16. The **Discrete Fourier Transformation** (DFT) decomposes a given discrete signal into the superposition of sine- and cosine-waves. The coefficients of the different waves are called the Fourier spectrum. A specific implementation of a DFT represents the spectrum S of a signal s with N components as 1P

$$S = [S(0), S(1), \dots, S(N/2), S(-N/2), S(-N/2+1), \dots, S(-2), S(-1)] \in \mathbb{C}^N$$

The application of this DFT implementation to a given, **real-valued signal** x with $N=17$ components results in the vector X , i.e. the **spectrum** of x . The first nine components of X are given below, while the last eight components are not explicitly calculated for computational efficiency.

$$X = [10+0i, -5+1i, -0.5+0.2i, -0.04+0.03i, 0, 0, 0, 0, 0, ?, ?, ?, ?, ?, ?, ?]$$

Which values have these **eight unknown components** (denoted by “?” above)?

- i) $X = [\dots, 0, 0, 0, 0, 0, -0.04-0.03i, -0.5-0.2i, -5-1i]$
 - ii) $X = [\dots, 0, 0, 0, 0, 0, +0.04+0.03i, +0.5+0.2i, +5+1i]$
 - iii) $X = [\dots, 0, 0, 0, 0, 0, +0.04-0.03i, +0.5-0.2i, +5-1i]$
 - iv) $X = [\dots, 0, 0, 0, 0, 0, -0.04+0.03i, -0.5+0.2i, -5+1i]$
17. Using the framework of Fourier descriptors, it is easily possible to **smooth** a given shape. The components of the spectrum $X \in \mathbb{C}^{22}$ of the original shape x are given below. 1P

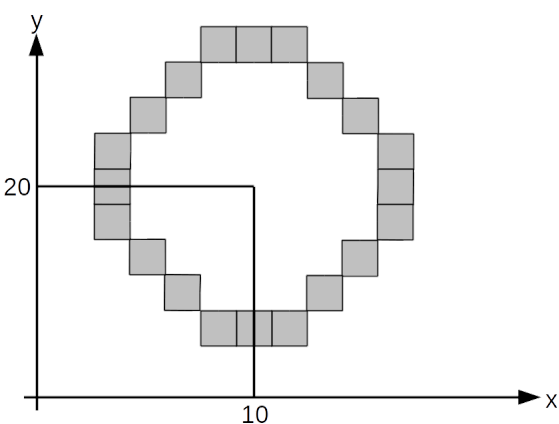
$$X = [130+50i, -30-4i, 3-12i, -9-4i, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -3-12i, -70+10i]$$

Which of the following spectra $Y \in \mathbb{C}^{22}$ would lead to a **smoothed version** y of x , if an **inverse DFT** is applied to them, i.e. $y = IDFT(Y)$.

- i) $[130+50i, -30-4i, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -3-12i, -70+10i]$
 - ii) $[139.3, 30.3, 12.4, 9.8, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 12.4, 70.7]$
 - iii) $[130+50i, -30-4i, 3-12i, -9-4i, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -3-12i, -70+10i]/30.3$
 - iv) $[130+50i, -22-3i, 1-3i, -0.5-0.2i, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -1-3i, -52+7i]$
18. The figure on the right shows a **circle** in a 2D coordinate system. Each of the grey squares corresponds to a single pixel. 2P

The outline of this shape is described by a **complex-valued vector** u , which contains the **x-coordinates** in the real-, the **y-coordinates** in the imaginary part.

What is the value of the **0-frequency** if u is transformed into the Fourier spectrum, i.e. $U = DFT(u)$?



i) $U(0) = 0 + 0i$	ii) $U(0) = 10 + 20i$
iii) $U(0) = 20 + 10i$	iv) $U(0) = 200 + 400i$

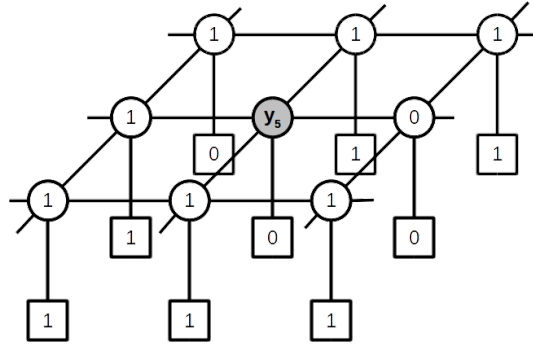
Block VII

19. Given three distinct events A, B, and C, where A and B are **conditional independent** given C. Which of the following statements is true? 1P

i) $P(A,B C) = P(A C)P(B C)$	ii) $P(A B,C) = P(A C)$
iii) $P(C A,B) = P(C A)$	iv) $P(A,B) = P(A)P(B)$

20. The figure on the right shows a small part of a specific **graphical model**. It consists of multiple nodes connected via undirected links.

This particular graphical model shall be used for **semantic segmentation** of a **binary image** $\mathbf{x} \in \{0,1\}^{N \times M}$ with $N \times M$ pixels. In particular, the goal is to estimate $P(\mathbf{y}|\mathbf{x})$ where $\mathbf{y} \in \{0,1\}^{N \times M}$ are the class labels of each pixel in the image.



The **measurement nodes** are denoted by squares and initialized with the given binary image. The **label nodes** are denoted by circles. The **unary energy** ψ_i is based on the result of a probabilistic classifier and given as $\psi_i = -\log(P(y_i|x_i)) - \log(e^1 + e^{-1})$ where $P(y_i|x_i) = \exp(I(y_i, x_i)) / (e^1 + e^{-1})$ and $I(y_i, x_i) = 1$ if $y_i = x_i$ and $I(y_i, x_i) = -1$ otherwise. The **binary energy** ϕ_{ij} is given as $\phi_{ij} = -(2y_i - 1)(2y_j - 1)$.

What is the **local energy** at node y_5 for $y_5=1$ given the graphical model depicted in the figure?

i) 1	ii) -1	iii) -2	iv) 2
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21. What is a **CRF**? 1P

i) Clustering Random Forest	ii) Classical Random Forest
iii) Conditional Random Field	iv) Canonical Random Field

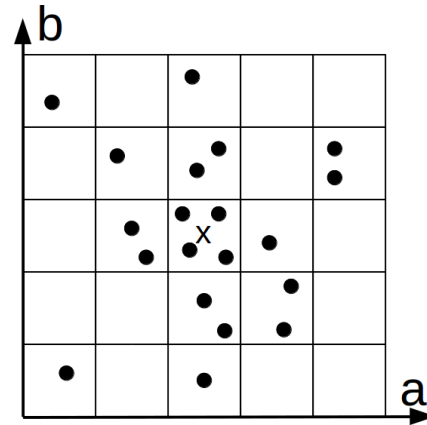
Block VIII

22. The figure on the right shows **samples** (denoted as black circles) in a **two-dimensional feature space**. The space is divided into a grid of square cells.

A **histogram** is used to estimate the probability distribution based on these samples and the shown grid.

A **Laplace correction** is applied with $\alpha = 1$.

The probability density of the sample at the position marked with an "x" in the figure is



2P

i) 1/4	ii) 1/9	Iii) 1/20	iv) 1/5
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23. Given one-dimensional samples $x \in [0,100) \subset \mathbb{N}$. Which of the following methods belongs to **parametric techniques** to estimate a **probability density** from these samples? 1P

- i) Model $P(x)$ as $P(x) \approx P(4i \leq x < 4(i+1)) = \theta_i$ with $0 \leq i < 25$ and estimate the parameters θ_i .
- ii) Histograms
- iii) Gaussian Mixture Models
- iv) Parzen Window

24. Two **probabilistic models** are given by their respective **parameters** Ω_1 and Ω_2 . The **likelihood of these models** on data D is $P(D|\Omega_1) = 2 \cdot 10^{-6}$ and $P(D|\Omega_2) = 10^{-6}$. Furthermore, it is known that $P(\Omega_1) = 1/4$ and $P(\Omega_2) = 3/4$. 2P

The **probability of a data sample** x based on the two models is given as $P(x|\Omega_1) = 1/4$ and $P(x|\Omega_2) = 1/3$.

Given the information above and using **maximum likelihood** for model selection, the estimate of $P(x|D)$ is

i) 1/4	ii) 1/3	iii) 2/5	iv) 3/10
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25. Given the information above and using a **MAP estimator** for model selection, the estimate of $P(x|D)$ is 2P

i) 1/4	ii) 1/3	iii) 2/5	iv) 3/10
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26. Given the information above and using **Bayesian parameter estimation** for model selection, the estimate of $P(x|D)$ is 3P

i) 1/4	ii) 1/3	iii) 2/5	iv) 3/10
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